



Tomato ripeness detection using hyperspectral imaging technology with multimodal fusion and improved 1D-CNN models

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Received: 8 December 2025 / Accepted: 2 March 2026 / Published online: 13 March 2026

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Abstract

Traditional tomato ripeness detection mostly relies on single-color or single-spectral technologies, which have limitations such as limited detection accuracy and susceptibility to environmental interference. Since tomato ripeness significantly affects its quality indicators, ripeness determination is crucial in the agricultural production and food processing industries. In this study, tomato spectral data and ripeness characterization data were collected, and Gaussian Process Regression (GPR), Partial Least Squares Regression (PLSR), and Support Vector Regression (SVR) were used to quantitatively characterize the spectral features associated with tomato ripeness. The optimal combination of preprocessing methods, feature wavelength extraction methods and modeling methods was selected through comparative experiments. Based on this optimal modeling scheme, hyperspectral feature wavelengths were fused with the hue (H) values in the HSV color space, and the detection performance of the One-Dimensional Convolutional Neural Network (1D-CNN) and machine learning methods was compared. In addition, the Convolutional Block Attention Module (CBAM) was incorporated to construct the 1D-CBAMCNN model for tomato ripeness detection, which achieved an accuracy of 95.18%, a 2-percentage-point improvement compared with the benchmark model 1D-CNN. Meanwhile, the SHapley Additive exPlanations (SHAP) method was adopted to analyze the importance of feature wavelengths, enhancing the interpretability of the model. This study provides a feasible solution for high-precision and interpretable non-destructive detection of tomato ripeness, and has practical significance for promoting the development of precision agriculture and post-harvest processing technologies.

Keywords Hyperspectral imaging · Multimodal fusion · Feature wavelength extraction · Tomato ripening

Introduction

Tomatoes, one of the three major globally traded vegetables with high nutritional and economic values, occupy an important position in vegetable cultivation, and their ripeness directly affects fruit quality, market value and supply chain management efficiency [1, 2]. Harvesting under-ripe tomato fruits may reduce the taste quality of the ripened fruits, and harvesting over-ripe tomatoes may lead to over-ripening of the fruits during export transshipment, which increases the risk of spoilage and results in significant losses in transportation and storage [3–5].

As the most intuitive external indicator of tomato ripeness, color-based detection technology is a research focus in fruit and vegetable quality evaluation, and computer vision-based methods have been widely used due to their convenience and high efficiency [6, 7]. Tomato ripening is also accompanied by internal quality changes, such as lycopene accumulation, sugar content increase, acidity reduction, and

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firmness decline caused by cell wall degradation. Meanwhile, hyperspectral imaging technology can capture subtle spectral differences to identify internal chemical composition and structural changes, thus compensating for the limitations of sole surface color-based detection [8, 9]. Tomato ripening is an inside-out process with asynchronous internal and external maturation, so a single technique often still has certain limitations; whether it relies on the external color features captured by computer vision, or only on internal structure and chemical composition information obtained by hyperspectral spectroscopy, it is difficult to comprehensively and accurately determine the ripeness of tomatoes [10]. The use of multimodal fusion technology to accurately assess the ripeness of tomato by integrating color, internal quality and firmness has become an effective way to break through the current dilemma [11].

Given the limitations of single-modal detection techniques mentioned above, integrating multi-source information through multimodal fusion has become a key direction to address this challenge, and the rapid development of artificial intelligence technology has provided powerful technical support for this approach. In particular, deep learning models have demonstrated prominent advantages in handling complex data and modeling, enabling more efficient integration and analysis of multi-source heterogeneous data such as color features and spectral information in tomato ripeness evaluation, thus holding the promise of achieving accurate determination of tomato maturity [12–14]. Recent studies have shown that architectures such as CNN-LSTM and Transformer-based hybrid neural networks exhibit remarkable advantages in extracting spatio-temporal correlated features from high-dimensional sequential data, providing diverse solutions for multi-source heterogeneous data processing [12]. The one-dimensional convolutional neural network, a typical deep learning algorithm, combined with spectral technology, has been successfully applied in many fields. For example, it has been used in the detection of meat freshness [15], the measurement of soluble solids content (SSC) in apples [16], and the determination of the contents of dry matter (DMC), SSC, and firmness in sweet potatoes [17]. In addition, in the field of ensemble learning, sophisticated strategies such as stacked ensembles, model averaging, and hybrid composite models have been proven to be more conducive to improving model robustness and accuracy compared with the basic ensemble algorithms adopted in this study, such as Random Forest and LightGBM [13]. Meanwhile, Gaussian Process Regression (GPR), as a powerful probabilistic nonparametric model, can provide uncertainty estimates while generating prediction results, and thus possesses unique application value in nonlinear regression tasks [18].

However, deep learning models are often regarded as “black boxes” when determining tomato maturity. Since they cannot clearly explain the basis and logic, the credibility of the models will be reduced [19]. To effectively solve the problem of interpretability, SHAPley Additive Explanations (SHAP), as a model-agnostic game theory method, has gradually attracted attention [20]. SHAP is able to generate feature attributes for each instance [21, 22]. Within the scope of chemometrics, SHAP values are of great value as they can elaborate in detail how different features, such as spectral wavelengths or chemical properties, affect the prediction results of the model [23]. In the interpretability analysis of complex data modeling, techniques such as partial dependence plots (PDPs), accumulated local effects (ALE), and network-based visualization methods have also been widely applied to elucidate model behaviors, providing diversified technical options for model interpretation [24, 25]. Meanwhile, the heterogeneity of internal and external ripeness during tomato ripening, combined with the limitation of poor interpretability of deep learning models, still greatly restricts the accurate determination of tomato ripeness. Although some scholars have attempted to overcome this dilemma through multimodal fusion technology, existing solutions have obvious shortcomings: most studies either fuse hyperspectral full-band data with color features or only rely on a single modality to optimize model performance. However, studies that specifically fuse screened hyperspectral characteristic wavelengths with more robust H values in the HSV space, use 1D-CNN enhanced by an attention mechanism to build a detection model, and simultaneously conduct systematic interpretability analysis of model decision-making using the SHAP method are still rare in this field. Based on this, this study aims to deepen the understanding of tomato ripeness, optimize the determination method, fill the aforementioned research gaps, and improve the accuracy and reliability of determination. Its main contributions are as follows:

1. Multimodal fusion data were constructed by fusing hue H-values using hyperspectral imaging technology, and the correlation between the characterization indexes and tomato ripeness was explored by comparing different spectral data preprocessing methods and feature wavelength screening methods, and by quantitatively analyzing the tomato's total soluble solids (TSS) and firmness (F) with models such as PLSR and support vector machine (SVM).
2. By comparing the ripeness detection results of machine learning and deep learning algorithms on single spectral data, single hue H-value, and spectral-color fusion data, we assessed the advantages and disadvantages of

different technological means in tomato ripeness monitoring, and finally established a tomato ripeness detection model based on 1D-CBAMCNN.

3. Aiming at the “black box” characteristics of deep learning models, the feature importance analysis of the tomato ripening classification model based on 1D-CBAMCNN was carried out using the SHAP method in order to reveal the influence of different spectral features on the accuracy of tomato ripening classification.

Dataset construction

Data collection

The tomato variety used in this study was Jingcai 8, a typical flavorful local cultivar in Beijing, which was collected from a plantation in Yanqing District, Beijing. After collection, all fruits were transported to the laboratory of Beijing Agricultural Technology Extension Station, Beijing, China. Upon arrival, the surface dirt of the tomatoes was wiped clean, and the fruits were then stored in a storage room at 20 °C for 12 h before experimental data collection on October 31, 2024. A total of 249 tomato fruits of uniform shape and size, without damage and free of pests and diseases, were selected. According to the industrial standard GH/T 1193–2021 of the China Supply and Marketing Cooperative Association, unripe tomatoes were not included in the formal classification due to their incomplete physiological development. All samples were divided into five ripening stages through manual labeling and automatic color feature quantification. Since the grouping was determined by the natural maturity distribution of the actual fruits, the number of samples in each stage was as follows: 50 green ripe stage (GRS) tomatoes, 49 color-changing stage (CCS) tomatoes, 50 early red ripe stage (ERR) tomatoes, 50 middle red ripe stage (MRR) tomatoes, and 50 late red ripe stage (LRR) tomatoes (Table 1 and Fig. 1).

Table 1 Tomato ripening criteria

Ripeness	Morphological characteristics
Unripe stage	Fruits and seeds are not fully developed and shaped
Green ripe stage	The fruit is shaped, and the surface of the fruit is glossy, changing from green to white-green
Color-changing stage	Reddish or yellow halo spots begin to appear around the umbilicus, and the reddish surface of the fruit is less than
Early red ripe stage	Fruit with red surface 10%~30%
Middle red ripe stage	Fruit with red surface 40%~60%
Late red ripe stage	Fruit with red surface 70%~100%

Hyperspectral image collection

A hyperspectral data acquisition system was established in this study (Fig. 2). Hyperspectral images of tomatoes were captured by the portable Specim IQ hyperspectral camera manufactured by Specim in Finland, with ARRILITE 750 Plus halogen lamps used as supplementary light sources on both sides. The halogen lamps were adjusted at an angle of approximately 60 ° relative to the vertical plane to ensure uniform illumination on the sample surface. The camera adopted Specim push-broom imaging technology and was a line-scanning device integrated with a spectral camera lens and a field-of-view camera lens, which can simultaneously acquire hyperspectral images and visible light images. It covers a full spectral range of 397 to 1000 nm with a wavelength interval of 3 nm, including a total of 204 wavelength points that span the visible light band of 400 to 760 nm and the short-wave near-infrared band of 760 to 1000 nm. The visible light band is sensitive to the pigment characteristics of tomato pericarp, while the short-wave near-infrared band can reflect the spectral absorption characteristics of internal fruit components such as moisture and sugar. In addition, the camera is equipped with a standardized line scan imaging function, which can stably capture two-dimensional images with a resolution of 512 × 512 pixels.

Before image acquisition, the hyperspectral camera was fixed on a support and adjusted to the position directly above the target tomatoes. The halogen lamps were calibrated to ensure sufficient and uniform illumination in the acquisition area, and a white reference panel was used to capture the illumination characteristics [26]. After the calibration of camera position and light sources, the focal length was manually adjusted for clear imaging, and the capture button was pressed to initiate scanning. During scanning, the camera captured and saved images in PNG format synchronously. After scanning, it automatically generated RGB visual images and spectral curves of the samples, and the retention of acquired data was determined according to the actual imaging quality. The camera has a built-in black-and-white correction preprocessing function, which can collect and store raw data, dark frame data and white reference data simultaneously. The raw hyperspectral data were converted into reflectance data normalized to the range of 0 to 1 through Eq. (1), which is shown as follows:

$$R = \frac{Raw - dark}{White - dark} \quad (1)$$

where R denotes calibrated tomato hyperspectral data, Raw denotes raw tomato hyperspectral data, dark denotes dark frame data, and White denotes white plate data.

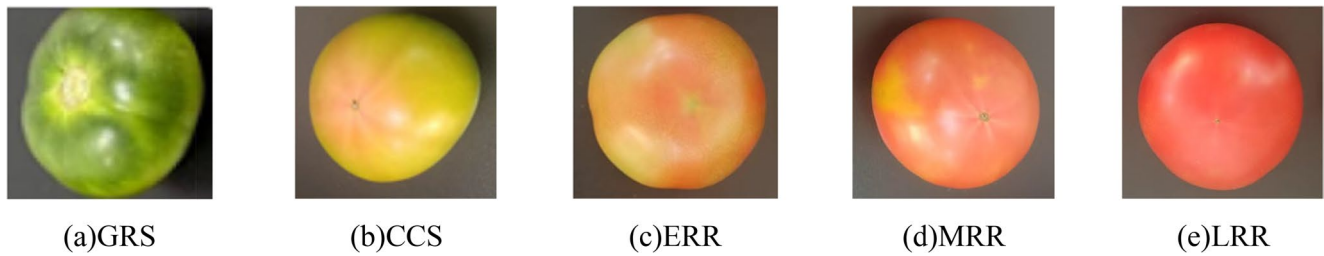
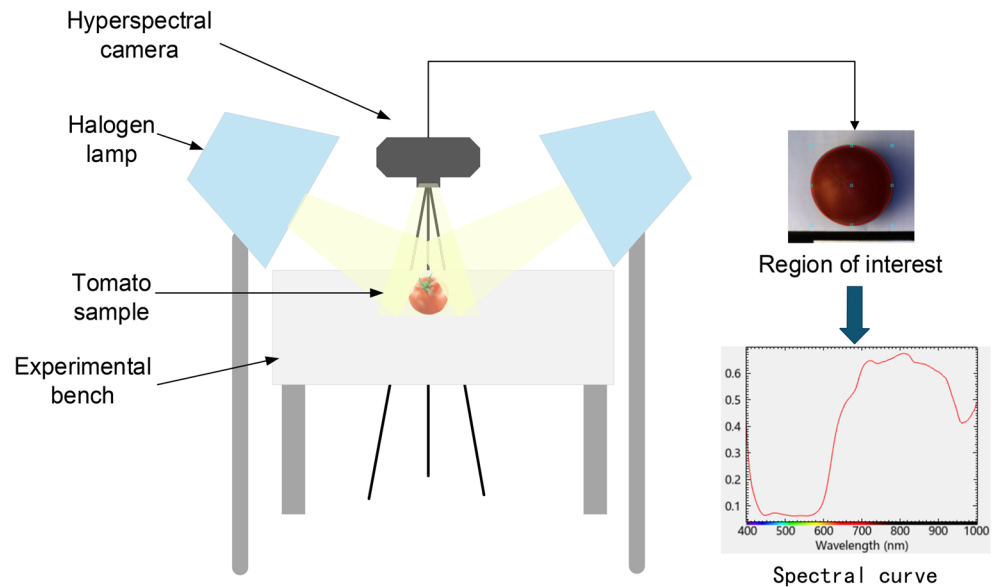


Fig. 1 Tomato samples of different ripeness (a)GRS (b)CCS (c)ERR (d)MRR (e)LRR

Fig. 2 Hyperspectral image collection



The camera saved an image in PNG format while collecting spectral data. After scanning, the RGB image captured by the hyperspectral camera and the spectral curve of each pixel can be displayed. The original hyperspectral image includes both tomato samples and background information. A circular region of interest (ROI) was manually selected on each tomato using ENVI 5.6 software. Tomatoes were placed with their calyx ends upward, the center of the circular ROI was set at the top of each tomato, and the edge of the circle was positioned near the equatorial region of the fruit. The average spectral data within the ROI was extracted for subsequent analysis.

Physical and chemical indicator data collection

Firmness is an important sensory index used to determine fruit ripeness [27]. Measurement of firmness was carried out using a GY-3 durometer. The probe of the durometer (3.5 mm in diameter) was inserted into the equatorial surface of the tomato fruits at a constant speed, and peeled test surfaces of about 1 cm² were prepared at the upper, middle and lower parts of the fruits, and the arithmetic

mean of the three measurements was taken as the final indicator of firmness.

TSS is the total amount of sugars, acids and other soluble substances in tomato pulp, and its content directly reflects the sweetness and flavor of tomato, which is one of the most important indexes for judging the ripeness of fruits [28, 29]. Measurement of TSS was carried out using a handheld brix meter (BM-04, Nohawk). Immediately before the measurement, the samples were squeezed out of the juice after removing the inedible portion and dripped into a digital refractometer for three measurements, and the average of these measurements was taken as the TSS measurement for tomatoes.

Data preprocessing

Hyperspectral data preprocessing

The collected spectral data primarily reflect the quality information of tomatoes, yet they inevitably contain irrelevant interference information such as background noise, electrical noise, baseline drift and scattering effects due to

the influence of detection environment, instrument performance and sample physical characteristics. These interferences will reduce the correlation between spectral data and tomato quality indicators, thereby impairing the accuracy and robustness of subsequent modeling [26]. Therefore, it is essential to preprocess and calibrate the raw spectral data before establishing prediction models. This step can eliminate interferences from non-target factors and enhance effective spectral characteristics, thereby ensuring the accuracy and reliability of spectral data and laying a solid foundation for subsequent model construction.

The preprocessing methods adopted in this study included Savitzky-Golay Smoothing (SG), Multiplicative Scatter Correction (MSC), Standard Normal Variate (SNV), and Mean Centering (MC), all of which have been extensively verified and applied in hyperspectral data processing for fruit and vegetable quality evaluation [26, 30, 31]. Specifically, SG smoothing can effectively reduce random electrical noise in the spectrum without distorting the spectral signal trend or altering the position and shape of characteristic peaks [30]. MSC and SNV are capable of eliminating scattering effects caused by uneven sample surfaces and particle size differences, while correcting baseline drift to improve the consistency of spectral data [30, 31]. MC reduces the impact of dimensional differences among variables through centering processing of spectral data, which further improves the efficiency and interpretability of subsequent modeling. These four methods work synergistically to effectively remove irrelevant interferences while completely preserving the key spectral characteristic peaks associated with the internal quality of tomatoes [31]. Preliminary comparative experiments were also conducted in the pre-experimental stage, where detrending and derivative transformation were tested alongside the selected methods. The results showed that detrending would lead to the loss of partial low-wavelength characteristic information, while derivative transformation would amplify high-frequency noise in the spectral data, thus compromising the stability of subsequent modeling.

Tomato surface color feature selection

Common color models used to describe the surface color of tomato are RGB, Lab, HSV, etc. The color component values, color moments, and color histograms of each color model are important features used to differentiate tomatoes of different ripeness levels [32, 33]. The png images corresponding to the spectral data acquired by the hyperspectral camera were removed and converted to the HSV color space to obtain color features that are more stable and independent of light changes [34, 35]. In this study, the function of Specim IQ camera that can simultaneously

capture hyperspectral and visible light images was utilized, and the raw color information was stored in RGB format. In order to realize the feature selection of tomato surface color, firstly, the tomato region was accurately keyed out by the image segmentation technique, using color threshold segmentation and edge detection to remove the background from the RGB image, where the RGB thresholds were set as $R > G > B$ (150–255, 30–120, 30–120) for red tomatoes, $G > R > B$ (80–200, 30–120, 30–120) for green tomatoes, $R/G/B < 20$ for black background and $R/G/B > 235$ for white background via experimental data processing. Then, the keyed tomato image was converted to HSV color model. The values of each pixel point in the H (hue), S (saturation), and V (lightness) channels were read, and the average of the values of the corresponding channels of all pixel points was calculated.

Materials and methods

The materials and methods mainly including the quantitative characterization of spectral features and detection of tomato ripening based on multimodal data fusion were as follows at Fig. 3. Step 1, data collection and pretreatment. Step 2, quantitative characterization and analysis of spectral features and detection of tomato ripening. Step 3, maturity grading of tomato ripening.

Quantitative characterization of spectral features of tomato ripening

Hyperspectral feature wavelength extraction

The spectral bands collected by the hyperspectral camera range from 400 to 1000 nm and contain 204 bands. Due to the presence of a large amount of environmental information in the raw spectral data that is not related to the internal composition of tomato, this information can affect the accuracy of ripening detection. In this study, Monte Carlo Uninformative Variable Elimination (MCUVE) and Variable Importance Projection (VIP) were used so as to eliminate the redundant and complex information in the raw wavelengths. The regression-based feature wavelength selection method was preferred over discriminant methods such as Analysis of Variance (ANOVA) because regression methods can directly establish a quantitative correlation between spectral features and internal quality indicators related to tomato ripeness, which aligns with the core objective of this study. In contrast, discriminant methods focus more on the significance of spectral differences between groups, making it difficult to capture the correspondence between spectral data and the continuous changes in tomato ripeness.

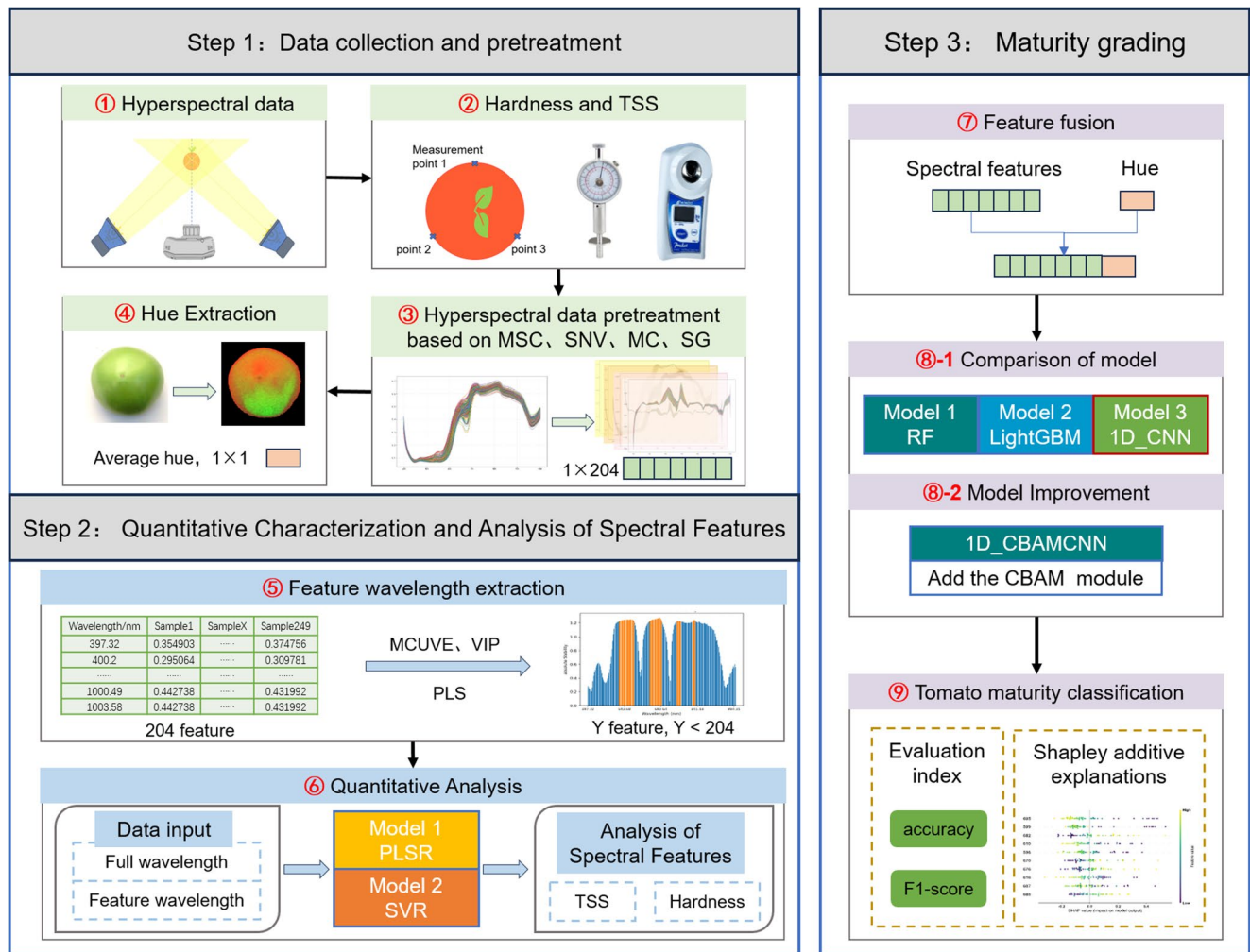


Fig. 3 Overview of the research process

Additionally, regression-based methods are more targeted at eliminating redundant information in high-dimensional spectral data and can effectively retain key wavelengths highly correlated with ripeness.

MCUIVE is a feature selection technique based on Monte Carlo method. Firstly, multiple PLSR models are constructed by randomly sampling the spectral matrix, then the regression coefficients of each model are calculated and the uninformative variable method is used to evaluate the importance of each variable to obtain the importance index I_i of each variable, and then the PLSR models of the corresponding variables are constructed sequentially by sorting the absolute value of I_i from the largest to the smallest and calculating the predicted root-mean-square error (RMSEV) of the models. Finally, the model with the smallest RMSEV is selected as the final model, and the last I_i added to this model is used as the threshold N . According to the threshold N , the wavelengths corresponding to all the variables with the absolute value greater than N are selected as the feature

wavelengths, and the variables corresponding to the wavelengths with the absolute value less than the threshold N are regarded as the uninformative variables and deleted. The calculation equation is as follows:

$$I_i = \frac{\text{mean}(w_i)}{\text{std}w_i} \quad (2)$$

where I_i denotes the absolute value of the significance indicator of the variable for the i th wavelength in the model; $\text{mean}(w_i)$ denotes the mean value of the regression coefficient w_i corresponding to the i th wavelength; $\text{std}w_i$ denotes the standard deviation of the regression coefficient w_i corresponding to the i th wavelength.

The VIP feature wavelength extraction algorithm is a feature selection method based on the PLSR model, where the VIP value measures the ability of each independent variable to explain the dependent variable, and when all the independent variables have the same contribution to the dependent

variable, all of them will have a VIP value of 1. Therefore, the higher the VIP value of a particular variable, the higher the contribution of the variable in explaining the dependent variable; and vice versa, the contribution will be smaller. In this study, when screening features, five experiments were executed using the VIP method, and the features with higher VIP values were selected as candidate variables by evaluating the contribution of each feature. Finally, by comparing the results of each experiment, the combination of features with the smallest root mean square error is selected as the final feature set. Its calculation equation is as follows:

$$VIP_t = \sqrt{\frac{1}{N} \sum_{j=1}^J \left(\frac{\partial Y_j}{\partial X_{tj}} \right)^2} \quad (3)$$

where X_{iw} denotes the value of the t th wavelength variable in the j th sample, Y_j denotes the value of the target variable in the j th sample, N is the total number of samples, and j is the number of scores.

Quantitative characterization analysis of spectral features

In this study, three algorithms including PLSR, SVR and Gaussian Process Regression (GPR) were selected to construct quantitative prediction models [18, 36]. A 5-fold cross-validation scheme was adopted for model training and performance evaluation to avoid accidental errors caused by a single data partition, thereby improving the reliability and statistical significance of the results (Table 2).

Grid search was employed to optimize the hyperparameters of the three models one by one, so as to achieve accurate prediction of the total soluble solids and firmness of tomatoes. The hyperparameters optimized for each model are listed as follows:

Evaluation indexes

For the quantitative evaluation of agricultural product physicochemical indicators, this study selected the coefficient of

determination (R^2), root mean square error (RMSE), relative root mean square error (RRMSE), and mean absolute percentage error (MAPE) as comprehensive evaluation criteria. These metrics are widely accepted and commonly used in regression tasks to evaluate model prediction performance. These indicators enable comprehensive performance comparison across different models and prediction targets, address the poor universality of indicator comparison in unbalanced datasets and multi-scale regression tasks, and enhance the interpretability of regression results. Notably, R^2 and RMSE are common metrics for regression analysis in this field, reflecting model goodness-of-fit and absolute prediction deviation respectively; RRMSE and MAPE, as relative error metrics, effectively eliminate dimensional and scale differences between prediction targets, with their application value validated in recent agricultural sensing studies [17, 37]. The calculation equations are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

$$PRMSE = \frac{RMSE}{\bar{y}} \times 100\% \quad (6)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \quad (7)$$

where y_i denotes the actual measured value, $\hat{y}_i$ is the predicted value, $\bar{y}$ denotes the average of the actual values, n denotes the number of samples.

Tomato ripeness detection based on multimodal data fusion

Data fusion and data segmentation

In this study, a feature-level fusion strategy was employed to fuse the selected hyperspectral wavelength data with hue (H) features from the HSV color space, thereby constructing a multimodal fusion feature set that organically integrates spectral and color information and lays a solid data foundation for subsequent analyses [38]. To improve the reliability and generalization ability of model evaluation, the experiment adopts a 5-fold cross-validation scheme for model training and performance verification. Specifically, the 249 tomato samples are randomly shuffled and evenly divided into 5 subsets. In each round of the experiment, 4 subsets

Table 2 Hyperparameters and search ranges for quantitative prediction models

Model	Hyperparameters to be optimized
PLSR	Number of latent variables (LVs): [1, 5, 10, 15, 20]
SVR	Penalty parameter C: [0.1, 1, 10, 100] Radial Basis Function (RBF) kernel parameter γ : [0.001, 0.01, 0.1, 1]
GPR	Insensitive loss parameter ϵ : [0.001, 0.01, 0.1, 0.2] Length-scale parameter of squared exponential kernel: [0.1, 1, 10] Observation model noise level: [0.001, 0.01, 0.1] Kernel function signal variance: [1, 10, 100]

are selected as the training set for model parameter learning, while the remaining 1 subset is used as the test set for evaluating model performance. The above process is repeated 5 times to ensure that each subset serves as the test set exactly once. By calculating the mean and standard deviation of the results from the 5 validation runs, the classification performance and stability of the model are comprehensively evaluated. This approach effectively avoids the accidental errors caused by a single data partition, making the experimental results more convincing and statistically significant.

1D-CNN model architecture

1D-CNN extracts local features by sliding with a one-dimensional convolutional kernel through the convolutional layer, the pooling layer compresses the features to downsize and retains the key information, and the fully-connected layer integrates the pooled features, and in this way, automatically extracts the data features for the final classification task. 1D-CNN usually takes a single spectral channel as the input when dealing with hyperspectral data, which makes it difficult for the model to capture the complex relationships between different bands. Due to the high dimensionality of hyperspectral data, 1D-CNN may ignore local detail information when extracting global features, which affects the discriminative ability of the model [39]. In addition, 1D-CNN is more dependent on the local features of the data, and it is difficult to fully explore the global contextual information in the image, which is particularly important in the tomato ripeness detection task, because the determination of ripeness is not only affected by local features, but also closely related to the overall morphological and color changes [40].

Based on the 1D-CNN architecture, the CBAM attention module is incorporated, containing three convolutional layers, three maximum pooling layers, two CBAM modules, and one fully connected layer [41]. The three one-dimensional convolutional layers sequentially perform hierarchical feature selection on the input spectral data, and in order to effectively reduce the computational complexity and enhance the generalization ability of the model, the maximum pooling layer is designed to reduce the dimensionality, as well as to reduce the number of parameters to avoid overfitting. Next, the extracted features are passed as inputs to the CBAM module, which weights

different feature channels and spatial locations by means of the channel attention mechanism (CAM) and spatial attention mechanism (SAM), so as to automatically filter out the most relevant and effective information for the classification task and assign appropriate weights to them. CAM generates channel weights via GAP, GMP, MLP and Sigmoid normalization, while SAM obtains spatial weights through channel-wise pooling, single-kernel convolution and Sigmoid. The 1D convolutional outputs first undergo CAM channel weighting, then SAM spatial weighting, forming a two-stage refinement to strengthen key features. Finally, after convolution and pooling operations, the features are flattened into one-dimensional vectors and fed into the fully connected layer for feature fusion and high-level nonlinear mapping, in order to finalize the classification task of spectral data (Fig. 4).

In order to improve the network training speed and mitigate overfitting, a batch normalization layer is introduced between the model convolution layer and the maximum pooling layer. Batch normalization reduces the sensitivity of the model to data distribution and improves the generalization ability of the model. The batch normalization for batch input data is calculated as:

$$y_b = \frac{1}{m} \sum_{i=1}^m x_i \quad (8)$$

$$f_b^2 = \frac{1}{m} \sum_{i=1}^m (x_i - y_b)^2 \quad (9)$$

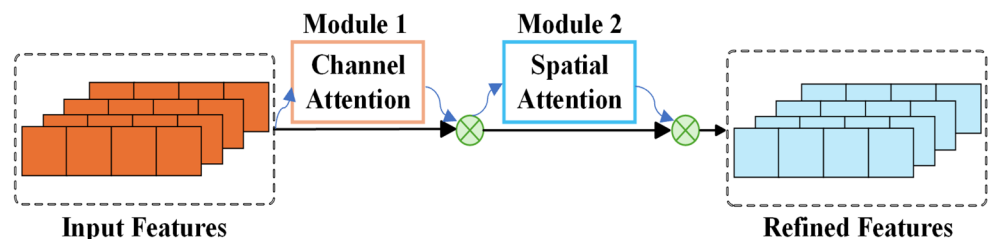
$$\hat{x}_i = \frac{x_i - y_b}{\sqrt{f_b^2 + \epsilon}} \quad (10)$$

$$s_i = \gamma \hat{x}_i + \beta \quad (11)$$

where y_b is the mean of the batch input data; f_b^2 is the variance of the batch input data; $\hat{x}_i$ is the normalized input; s_i is the output data after scale transformation and variance; and γ and β are the parameters that need to be learned when training the network.

The structure of the improved 1D-CBAMCNN model is shown in Fig. 5. The inputs of the model are 1D spectral data and 1D tomato hue H values. Since the original spectrum of tomato has 204 bands and 205 feature vectors after concurrent splicing with the tomato hue H features, the

Fig. 4 The CBAM module



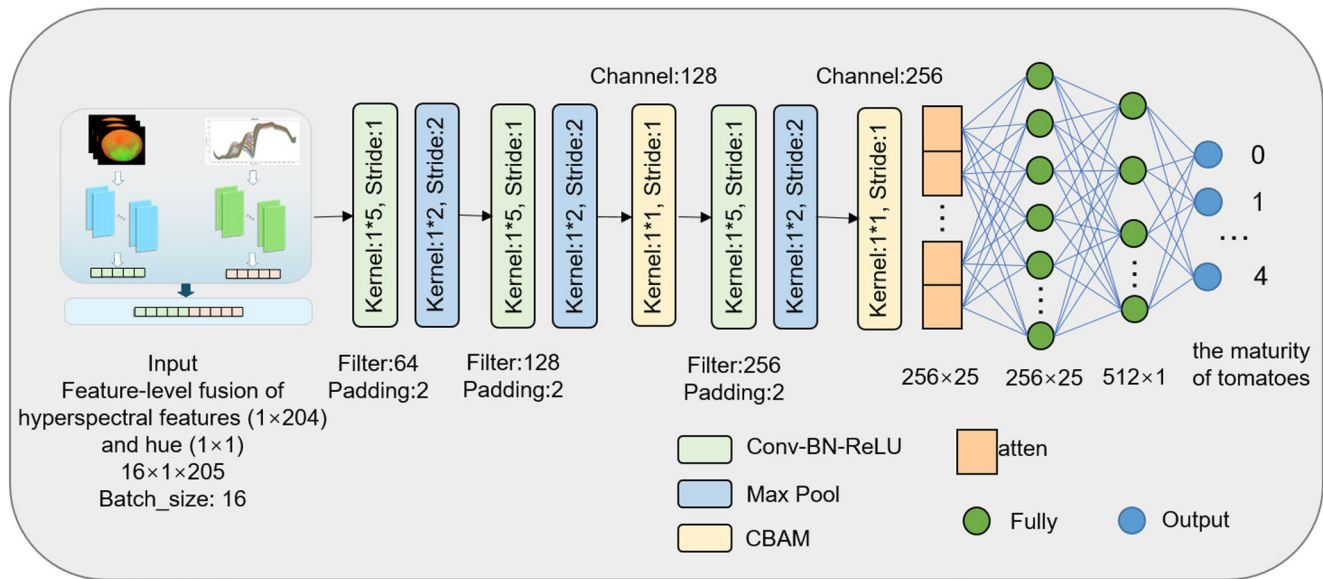


Fig. 5 1D-CBAMCNN network structure diagram

inputs of the model are the data with (batch-size, 1,205). batch-size is set to 16, and convolution kernels of size 1×5 are used in the convolution layers to extract the 1D spectral data and hue H features, and the number of convolution kernels in the three convolution layers are 64, 128 and 256, respectively. data and features of hue H. The number of convolution kernels in the three convolution layers is 64, 128 and 256, respectively, and the padding parameter in the convolution layer is set to 2 in order to avoid that the edge information is discarded due to the sliding of the convolution kernel in the process of convolution operation, which results in the loss of the information; the size of the pooling layer convolution kernel is 1×2 , and the step size is 2, and the CBAM is added after the second convolutional layer and the third convolutional layer respectively. module, the number of CBAM-1 channels is 128, and the number of CBAM-2 channels is 256, and the CBAM module enhances the feature importance by weighting the channel and spatial dimensions through the channel attention mechanism and spatial attention mechanism, respectively [42, 43].

During model training, the cross-entropy loss function is used as the loss function for the classification problem, and Adam is used as the optimizer to update the parameters of the model with a weight decay factor of 0.00001. The initial value of the learning rate is set to 0.001 and the learning rate is updated using the cosine annealing learning rate scheduler with an activation function of Relu, a Batch-size of 16, and a Dropout rate is 0.5, and the maximum number of training iterations is 300.

Hyperparameter tuning for the remaining comparative models

RF is a typical Bagging-based ensemble model using decision trees as base learners. It generates subsets via bootstrap sampling, trains trees with random feature selection, outputs results by voting or averaging, and effectively reduces overfitting risks with strong robustness [44]. LightGBM is a Boosting-based ensemble model optimized with a histogram algorithm. It discretizes continuous features to accelerate split point search, cuts down memory and computational costs significantly, and is ideal for large-scale high-dimensional data scenarios [45]. XGBoost is a Boosting-based ensemble model improved on gradient boosting decision trees. It trains trees sequentially by fitting residuals, uses L1/L2 regularization to avoid overfitting, supports missing value handling, and works well for high-dimensional small-sample spectral data analysis [46]. CatBoost is a Boosting-based ensemble model, notable for its compatibility with categorical features. It adopts ordered boosting to resolve gradient bias, processes categorical features without complex encoding, simplifies multimodal data fusion, and features fast training and easy tuning [47]. MLP is a basic fully connected deep learning model consisting of input, hidden and output layers. It fits nonlinear data relationships via activation functions, optimizes parameters through backpropagation, and is often used as a baseline model for spectral prediction tasks [48]. Transformer is a self-attention-based deep learning model. It enables parallel

training, captures long-range feature dependencies, adaptively assigns weights to multimodal features, and performs excellently in tomato quality prediction with fused spectral and color features [49]. The hyperparameters optimized for each model are listed as follows: (Table 3).

Evaluation metrics

In order to comprehensively evaluate the performance of the tomato maturity classification model, this study adopts Accuracy and F1-score as the core evaluation metrics, and visualizes the classification details through Confusion Matrix [36].

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (12)$$

$$Precision = \frac{TP}{TP+FP} \quad (13)$$

$$Recall = \frac{TP}{TP+FN} \quad (14)$$

Table 3 Hyperparameters and search ranges for the comparative models of tomato ripeness detection

Model	Hyperparameters to be optimized	Model	Hyperparameters to be optimized
RF	n-estimators: [30, 50, 90, 120], max-depth: [10, 15, 20], min-samples-split: [2, 3, 5], min-samples-leaf: [1, 2], max-features: [0.5, 0.8]	XGBoost	n-estimators: [50, 70, 100, 120], learning-rate: [0.05, 0.1], max-depth: [4, 5, 6], subsample: [0.6, 0.7, 0.8], colsample-bytree: [0.7, 0.8, 0.9, 1.0]
LightGBM	n-estimators: [100, 200, 300, 400], learning-rate: [0.01, 0.05, 0.1, 0.2], max-depth: [2, 4, 6, 8, 10], num-leaves: [20, 30, 40, 50, 60]	CatBoost	iterations: [200, 300, 400, 500], learning-rate: [0.05, 0.1, 0.15, 0.2], depth: [1, 2, 3, 4, 5, 6, 7, 8, 9]
Transformer	batch-size: [8, 16], d-model: [16, 32], nhead: [2, 4], num-layers: [1, 2, 3], dim-feedforward: [32, 64, 128], dropout: [0.1, 0.2], learning-rate: [0.001, 0.0005], weight-decay: [0.0005, 0.001]	MLP	batch-size: [8, 16], hidden1: [32, 64], hidden2: [16, 32], dropout: [0.1, 0.2], learning-rate: [0.001, 0.0005], weight-decay: [0.0005, 0.001]

$$F1 - Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (15)$$

where TP denotes the number of samples correctly predicted to be in the positive category. TN denotes the number of samples correctly predicted to be in the negative category. FP denotes the number of samples in the negative category that were incorrectly predicted to be in the positive category. FN denotes the number of samples in the positive category that were incorrectly predicted to be in the negative category.

Experimental environment

All model training and testing experiments were conducted on a workstation with the following hardware specifications: Intel® Core™ i9-13900 K CPU (24 cores, 32 threads, base frequency 3.0 GHz, maximum turbo frequency 5.8 GHz), NVIDIA GeForce RTX 4090 GPU (24 GB GDDR6X video memory, 16384 CUDA cores), 64 GB DDR5 5600 MHz memory, and 2 TB NVMe SSD for fast data reading and model storage. The software environment was built on Windows 11 Professional 64-bit operating system, with PyTorch 2.1.0 (compatible with CUDA 12.1) as the deep learning framework and Python 3.9.16 as the programming language. Key dependencies included NumPy 1.24.3, Pandas 2.0.3, Scikit-learn 1.2.2, OpenCV 4.8.0, ENVI 5.6, SHAP 0.42.1, and Matplotlib 3.7.1, which supported numerical computation, data processing, model implementation, image processing, hyperspectral preprocessing, interpretability analysis, and result visualization respectively.

Result

Analysis of Hue

As shown in Fig. 6, it is a comparative schematic diagram of the same tomato image with different brightness levels under the HSV color model. Through the color jittering method in data augmentation, the brightness differences caused by different natural environments and acquisition devices in real-world scenarios are simulated. It can be seen that for the same tomato image under different brightness levels, the H channel is minimally affected by brightness, the S channel is affected to a moderate extent, and the V channel is significantly affected. The H channel can effectively reduce the impact of brightness on the image effect. As can be seen from Fig. 7, the hue H value of the tomato gradually decreases from 39.59° during the green ripe stage to 4.60° in the late red ripe stage, showing the pattern that the hue H value continuously decreases as the maturity increases. This change indicates that the surface color of the tomato gradually transitions from green to orange-yellow and ultimately tends towards red. Specifically, the hue

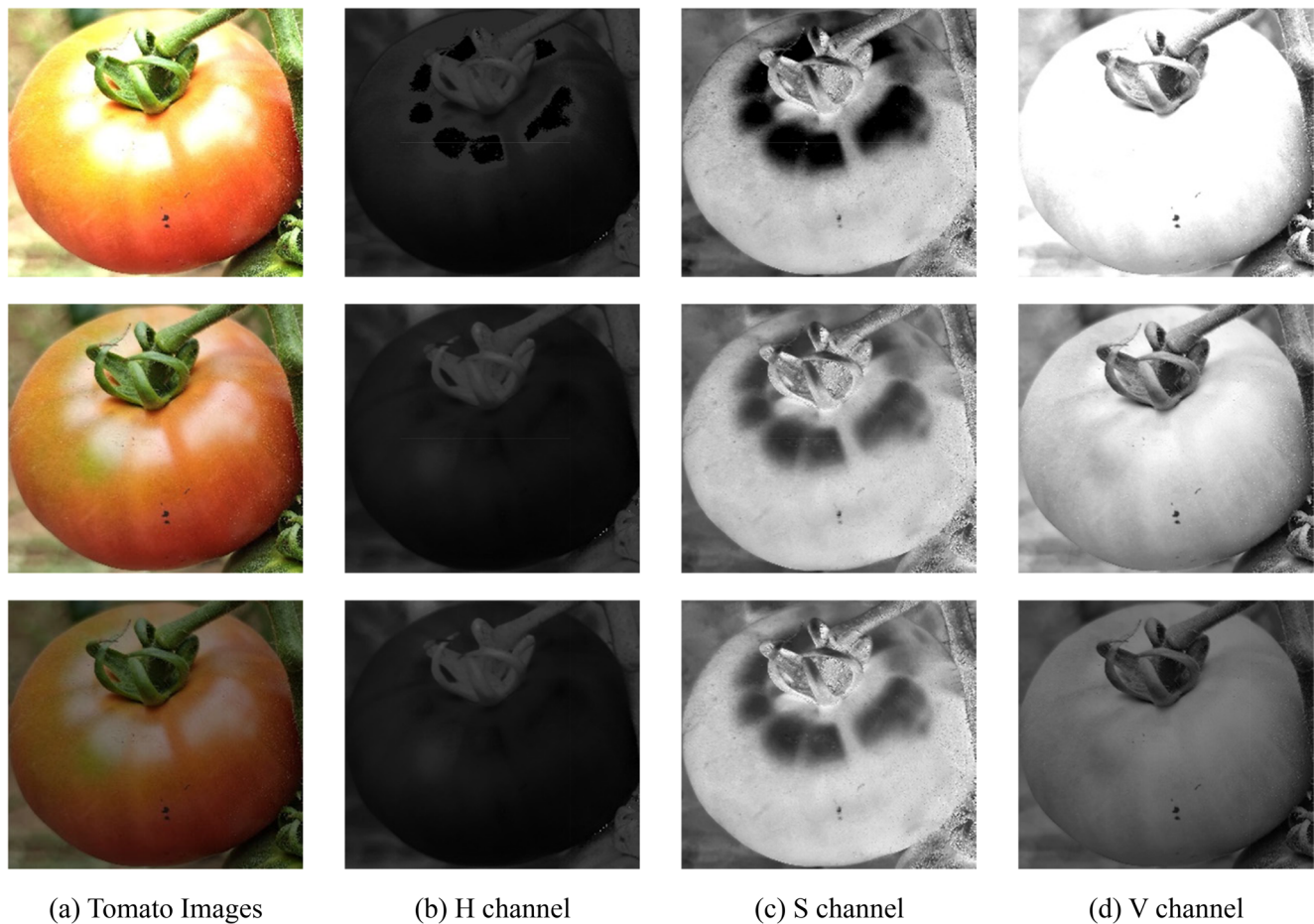


Fig. 6 Schematic comparison of the same tomato image with different luminance under HSV color model. **(a)** Tomato Images, **(b)** H channel, **(c)** S channel, **(d)** V channel

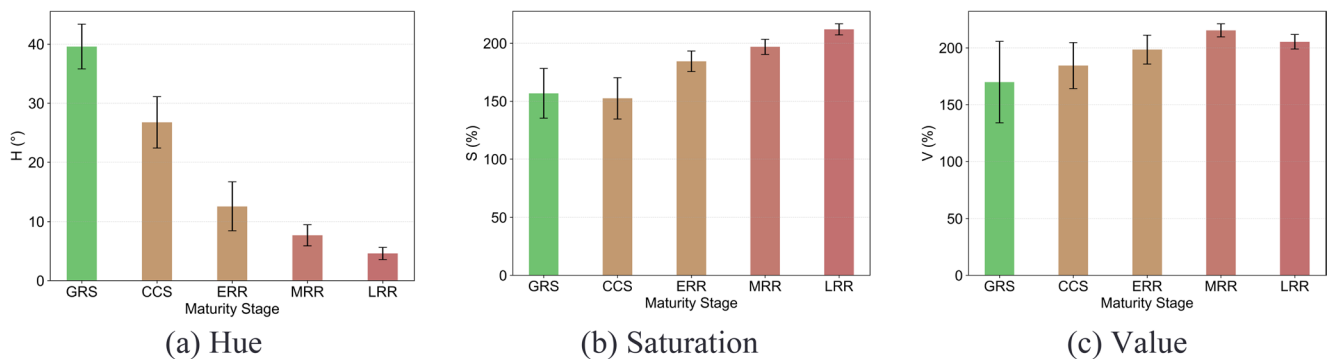


Fig. 7 Color characteristics of different ripe tomatoes. **(a)** Hue, **(b)** Saturation, **(c)** Value

H values during the green ripe stage, color transition stage, and early red ripe stage are 39.59°, 26.75°, and 12.57° respectively, with a relatively large variation range of approximately 14°, indicating that the color of the tomato rapidly changes from green to orange-red during this stage. In the mid-red ripe stage and late red ripe stage, the hue H values drop to 7.68° and 4.60° respectively, and the values are relatively close, indicating that

the color of the tomato is basically stable and presents a red hue during this stage. Based on the comprehensive analysis, it can be known that the changing trend of the hue H value is highly consistent with the ripening process of the tomato, and it is less affected by brightness interference. The value of the H channel can be used as an important indicator to characterize the ripeness of the tomato.

Spectral characterization

The initial raw spectra of all samples ($n = 249$) are shown in Fig. 8(a), and the average spectral profiles of tomatoes at five different ripening stages are shown in Fig. 8(b). The wavelength with the maximum reflectance was located in the range of 820–830 nm, and the corresponding reflectance value ranged from 0.57 to 0.70. The ripening of tomato is manifested by significant changes in its internal nutrient composition [50]. These changes in physicochemical properties are reflected in the spectral reflectance curves.

The curves show the differences in spectral reflectance of tomatoes at different stages of ripening in the range of 390–1004 nm. The spectral differences in the 500–600 nm and 600–720 nm bands were significant, and during tomato ripening, the spectral reflectance in the 500–600 nm range decreased while the spectral reflectance in the 600–720 nm range increased, which was attributed to the gradual change of the color from green to red during the ripening process of the tomato fruit [51]. This is due to the gradual color change from green to red during tomato fruit ripening. Spectral reflectance near 670 nm is related to chlorophyll

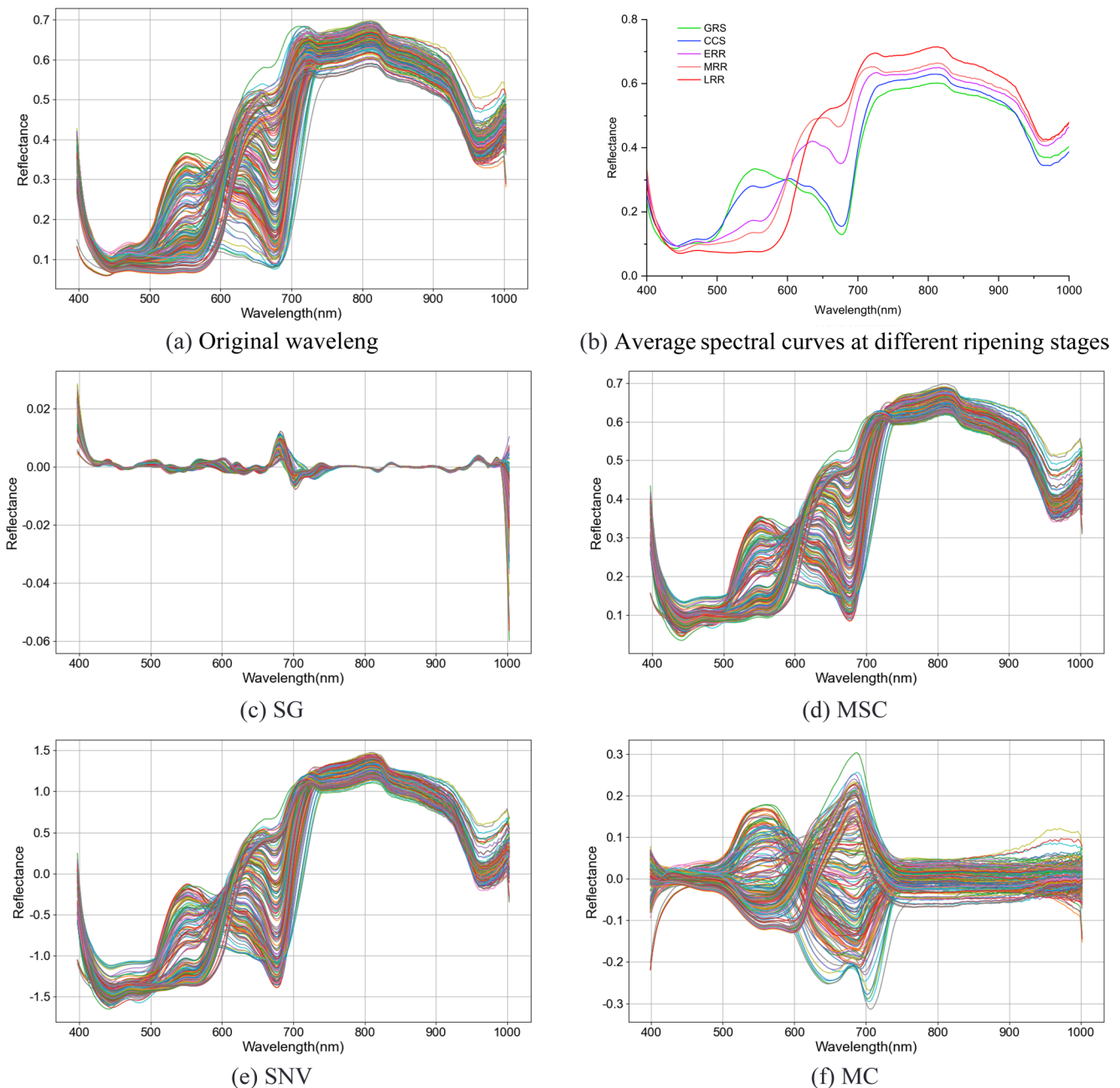


Fig. 8 Spectral pre-processing. (a) Original wavelength, (b) Average spectral curves at different ripening stages, (c) SG, (d) MSC, (e) SNV, (f) MC

absorption in tomato, which decreases or completely disappears when tomato ripens from green to red, so the spectral reflectance at 660 and 685 nm steadily increases [52]. Spectral reflectance in the 500–600 nm band is related to carotenoids, so the spectral reflectance at 552 and 561 nm steadily decreases [52]. The spectral reflectance at 552 and 561 nm steadily decreases [53]. The reflectance at different maturation stages between 700 and 800 nm gradually increases until the highest value near 820 nm, and shows a decreasing trend from 820 nm to 950 nm until there is a clear absorption peak at 950 nm, which is mainly affected by the O-H bond [54]. During tomato ripening, the decrease in chlorophyll content and the increase in lycopene together lead to changes in the tomato reflectance spectral profile. These changes indicate a clear correlation between the optical properties of tomato and its natural growth process.

In spectral analysis, the collected spectral images are easily affected by external ambient light, measurement conditions, instrument performance and other factors, so the original spectral information contains not only the information of the measured sample, but also irrelevant information such as the sample background and the calibration status of

the instrument, Fig. It can be seen that the spectral curve after 800 nm is slightly perturbed. In order to ensure the accuracy and reliability of the spectral data, the spectra were pre-processed with SG, MSC, SNV, and MC to minimize the influence of external interference. The pre-processed spectral data are shown in Fig. 8(c)-(f).

Results of quantitative characterization analysis of spectral features of tomato ripening

Quantitative analysis results based on full wavelengths

The original spectra of tomatoes at different ripeness levels and the preprocessed spectra were used as inputs. The spectral data of the total 249 tomato samples were subjected to classical five-fold cross-validation and then fed into the quantitative analysis algorithms (GPR, PLSR and SVR) for the determination of TSS content and firmness.

Table 4 presents the full-wavelength quantitative analysis results of tomato TSS content and firmness, showing that spectral preprocessing methods significantly affect the performance of GPR, PLSR, and SVR models, and the

Table 4 Results of full-wavelength quantitative analysis

Indicators	Model	Preprocessing methods	R^2	RMSE	RRMSE / %	MAPE / %
TSS	GPR	unprocessed	0.506±0.131	0.564±0.080	8.121±1.128	5.811±0.616
		MSC	0.587±0.062	0.520±0.072	7.485±0.984	5.440±0.515
		SNV	0.549±0.051	0.545±0.078	7.848±1.061	5.703±0.625
		SG	0.579±0.071	0.524±0.069	7.540±0.960	5.436±0.569
		MC	0.543±0.075	0.548±0.088	7.891±1.214	5.687±0.685
	PLSR	unprocessed	0.546±0.058	0.544±0.060	7.837±0.794	5.734±0.323
		MSC	0.576±0.056	0.527±0.062	7.582±0.844	5.467±0.510
		SNV	0.583±0.066	0.522±0.065	7.510±0.891	5.490±0.477
		SG	0.556±0.059	0.539±0.069	7.765±0.941	5.785±0.558
		MC	0.549±0.062	0.544±0.077	7.828±1.043	5.666±0.535
	SVR	unprocessed	0.541±0.078	0.549±0.084	7.897±1.148	5.545±0.545
		MSC	0.573±0.068	0.529±0.078	7.618±1.073	5.401±0.602
		SNV	0.590±0.059	0.519±0.079	7.473±0.988	5.416±0.528
		SG	0.588±0.063	0.519±0.072	7.475±1.083	5.462±0.635
		MC	0.559±0.048	0.538±0.071	7.748±0.955	5.552±0.503
Firmness	GPR	unprocessed	0.733±0.052	1.677±0.081	30.558±3.260	30.558±3.260
		MSC	0.739±0.040	1.667±0.141	30.350±3.464	25.454±5.562
		SNV	0.747±0.036	1.637±0.089	29.797±2.798	25.461±5.488
		SG	0.742±0.049	1.644±0.038	29.924±2.542	25.344±5.223
		MC	0.738±0.047	1.661±0.041	30.220±2.291	24.955±4.108
	PLSR	unprocessed	0.700±0.044	1.781±0.057	32.411±2.614	32.411±2.614
		MSC	0.703±0.075	1.752±0.151	31.773±2.404	27.809±3.089
		SNV	0.752±0.037	1.622±0.087	29.507±2.721	25.175±5.268
		SG	0.706±0.043	1.764±0.034	32.057±1.879	27.961±3.751
		MC	0.702±0.053	1.773±0.086	32.297±3.282	29.261±5.206
	SVR	unprocessed	0.712±0.039	1.761±0.237	31.984±4.353	24.998±3.506
		MSC	0.714±0.041	1.745±0.122	31.679±2.474	25.562±3.576
		SNV	0.747±0.055	1.624±0.109	29.550±2.910	24.276±2.516
		SG	0.637±0.072	1.952±0.080	35.481±2.599	27.044±4.429
		MC	0.737±0.057	1.667±0.167	30.381±4.098	26.324±6.554

prediction effect of firmness is generally better than that of TSS content. For TSS content prediction, the SVR model combined with SNV preprocessing achieves the optimal performance, with the highest R^2 (0.590 ± 0.059) and the lowest RMSE (0.519 ± 0.079) and RRMSE ($7.473 \pm 0.988\%$), while the SG-preprocessed GPR model and MSC-preprocessed PLSR model also show relatively good results, with R^2 values exceeding 0.57 and lower error indicators. For firmness prediction, the PLSR model with SNV preprocessing outperforms other combinations, achieving the highest R^2 (0.752 ± 0.037) and the lowest RMSE (1.622 ± 0.087) and RRMSE ($29.507 \pm 2.721\%$), followed by the SNV-preprocessed GPR and SVR models with R^2 values of 0.747 ± 0.036 and 0.747 ± 0.055 respectively. In contrast, all models without preprocessing have lower R^2 and higher error indicators, and the MC preprocessing method shows no obvious advantages in improving model performance for both TSS and firmness prediction. Overall, SNV preprocessing shows better performance in eliminating spectral noise for this study's tomato samples and scattering interference, thereby enhancing the predictive ability of the models for tomato internal quality indicators.

Results of feature wavelength extraction

The best preprocessing method plus model SNV-SVR for quantitative analysis at full wavelength of tomato TSS and firmness was selected, and two algorithms, MCUVE and VIP, were used to screen the feature wavelengths and establish a quantitative analysis model. The best subset of features was determined by minimizing the RMSE, and the number of features corresponding to this subset was used as the number of feature wavelengths selected.

Table 5 presents the quantitative analysis results of TSS content and firmness after screening by different feature wavelength extraction methods under SNV preprocessing, revealing the impact of feature wavelength selection on model performance. For TSS prediction with the SNV-SVR combination, the MCUVE algorithm (169

feature wavelengths) achieved the optimal performance ($R^2=0.593 \pm 0.065$, $RMSE=0.516 \pm 0.071$), slightly outperforming full-wavelength modeling (204 bands). The VIP algorithm retained only 49 feature wavelengths while maintaining comparable accuracy ($R^2=0.591 \pm 0.068$), simplifying the model effectively. In contrast, CARS (34 bands) and SPA (10 bands) led to decreased R^2 (<0.578) and increased errors due to excessive feature reduction. For firmness prediction using SNV-PLSR, the VIP algorithm (47 feature wavelengths) performed best ($R^2=0.762 \pm 0.051$, $RMSE=1.576 \pm 0.100$, $RRMSE=28.652 \pm 2.573\%$), significantly surpassing full-wavelength results. The MCUVE algorithm (77 bands) followed closely ($R^2=0.758 \pm 0.049$) with the lowest MAPE ($22.768 \pm 3.226\%$). Again, CARS and SPA with fewer feature wavelengths resulted in reduced prediction accuracy. Overall, MCUVE and VIP algorithms excel in feature wavelength screening, effectively eliminating redundancy, simplifying model complexity, and maintaining or improving accuracy. Excessively reducing feature wavelengths degrades performance, and the prediction effect of firmness is generally better than that of TSS content.

Results of tomato ripening detection based on multimodal data fusion

Results of classification experiments with different modalities

Based on the previously presented correlation analysis of tomato ripeness characterization indicators, ripeness is significantly correlated with the content of internal components such as TSS and fruit firmness, with fruit firmness showing the strongest association. Considering the natural imbalance in tomato ripeness grading and the susceptibility to bias in grading results based solely on color feature quantification, this study constructs a feature selection method by taking fruit firmness, TSS content, and ripeness classification labels annotated with color feature vectors as joint outputs.

Table 5 Results of quantitative analysis

Indicators	Model	Optimal preprocessing method	Feature wavelength extraction	Number of feature wavelengths	R^2	RMSE	RRMSE / %	MAPE / %
TSS	SVR	SNV	Original	204	0.590 ± 0.059	0.519 ± 0.079	7.473 ± 0.988	5.416 ± 0.528
			MCUVE	169	0.593 ± 0.065	0.516 ± 0.071	7.423 ± 0.977	5.405 ± 0.558
			VIP	49	0.591 ± 0.068	0.517 ± 0.072	7.445 ± 0.993	5.414 ± 0.492
			CARS	34	0.578 ± 0.070	0.526 ± 0.081	7.574 ± 1.118	5.427 ± 0.604
			SPA	10	0.574 ± 0.069	0.528 ± 0.077	7.606 ± 1.055	5.414 ± 0.547
Firmness	PLSR	SNV	Original	204	0.752 ± 0.037	1.622 ± 0.087	29.507 ± 2.721	25.175 ± 5.268
			MCUVE	77	0.758 ± 0.049	1.595 ± 0.108	29.022 ± 2.941	22.768 ± 3.226
			VIP	47	0.762 ± 0.051	1.576 ± 0.100	28.652 ± 2.573	26.319 ± 2.840
			CARS	34	0.732 ± 0.066	1.677 ± 0.194	30.451 ± 3.561	23.335 ± 3.157
			SPA	10	0.727 ± 0.079	1.688 ± 0.216	30.787 ± 5.016	26.558 ± 6.863

Table 6 Performance Comparison of Tomato Ripeness Detection Under Different Modal Data and Models

Different modal data	Model	Number of feature wavelengths	Accuracy/%	Precision/%	Recall/%	F1-Score/%
Spectral	1D-CNN	204	92.76±3.03	92.58±3.07	93.00±2.89	92.29±3.25
	LightGBM	53	90.75±3.40	90.99±3.32	91.01±3.01	90.4±3.52
	RF	53	90.39±3.33	90.92±3.25	90.7±3.35	90.16±3.64
Hue	1D-CNN	1	74.0±0.37	75.19±0.68	75.7±1.14	73.98±0.77
	LightGBM	1	75.19±1.35	76.34±0.83	76.84±1.32	75.22±1.36
	RF	1	77.61±3.17	78.28±2.78	78.74±2.57	77.51±2.38
Spectral + Hue	1D-CNN	205	93.56±3.03	93.65±3.05	93.66±2.91	93.18±3.39
	LightGBM	54	92.36±1.75	92.59±1.64	92.59±1.64	91.84±1.76
	RF	54	92.38±1.77	92.82±1.81	92.31±1.27	91.94±1.83

The determination of preprocessing and feature selection schemes focuses on enhancing firmness-related features: SNV, which optimally preserves firmness characteristics, is selected as the basic preprocessing method; VIP is adopted for feature selection, primarily because fruit firmness has the highest correlation with ripeness, enabling precise identification of spectral information strongly associated with core indicators.

Targeting the characteristics of different models, this study adopts a differentiated processing strategy: for machine learning models, spectral data are first preprocessed with SNV to eliminate spectral scattering interference and baseline drift, followed by screening 53 key feature wavelengths via the VIP algorithm. This simplifies the model structure, reduces computational load, and strengthens the support of firmness-related features for modeling. For deep learning models, leveraging their powerful ability to autonomously mine deep-seated features, no additional VIP feature selection is required—spectral data are directly input after SNV preprocessing to retain core information related to ripeness.

The comparison results of tomato ripeness detection performance under different modal data and models are shown in Table 6. It clearly demonstrates that spectral features are far more effective than single hue features for tomato ripeness detection, and multimodal fusion of spectral and hue data can further enhance model performance across the board. Specifically, in the single spectral modality, the 1D-CNN model achieves the highest accuracy of 92.76% without manual feature screening, outperforming the LightGBM (90.75%) and RF (90.39%) models that rely on 53

selected feature wavelengths. In contrast, all models perform poorly when using only hue data, with the best-performing RF model reaching just 77.61% accuracy. When spectral and hue features are fused, all models see accuracy gains: the 1D-CNN model's accuracy rises to 93.56%, while LightGBM and RF reach 92.36% and 92.38% respectively. This confirms that deep learning models have inherent advantages in handling high-dimensional spectral data and fusing multimodal information, and multimodal fusion is an effective way to improve the accuracy of tomato ripeness detection.

Comparative experiments of different models

As shown in Table 7, the 1D-CNN model outperforms all other models in comprehensive classification performance for tomato ripeness detection. Specifically, it achieves the highest accuracy of 93.18%, along with the optimal precision, recall and F1-score, which are 93.56%, 93.65% and 93.66% respectively. Unlike traditional machine learning models (RF, XGBoost, LightGBM, CatBoost) that rely on 54 manually screened feature wavelengths, the 1D-CNN model directly takes 205-dimensional multimodal fusion features (spectral+hue) as input, and autonomously extracts deep-level feature correlations through end-to-end learning, thus breaking through the performance bottleneck caused by manual feature engineering limitations. Although the 1D-CNN model has higher FLOPs (16.03 M) and longer average inference time per sample (2.04 ms) compared with traditional lightweight machine learning

Table 7 Performance comparison of tomato ripeness detection models

Model	Number of feature wavelengths	Accuracy/%	Precision/%	Recall/%	F1-Score/%	FLOPs/M	Average Inference Time per Sample/ms
RF	54	91.94±1.83	92.38±1.77	92.82±1.81	92.31±1.27	0.35	0.42
XGBoost	54	91.47±1.81	91.96±1.49	92.3±1.67	92.05±1.73	0.48	0.55
LightGBM	54	91.84±1.76	92.36±1.75	92.59±1.64	92.59±1.64	0.41	0.50
CatBoost	54	91.08±1.32	91.56±0.98	91.97±1.21	91.62±1.15	0.52	0.63
MLP	205	91.53±1.93	91.96±1.49	92.3±1.74	92.08±1.5	3.83	1.04
Transformer	205	92.01±2.84	92.36±2.25	92.63±2.39	92.68±2.57	9.61	1.52
1D-CNN	205	93.18±3.39	93.56±3.03	93.65±3.05	93.66±2.91	16.03	2.04

models, its computational complexity and inference speed are still within the acceptable range for practical deployment, especially for the field of agricultural on-site detection. In contrast, the MLP and Transformer models, which also use 205-dimensional features, show inferior classification performance to the 1D-CNN model, indicating that the 1D-CNN has a more suitable network structure for processing spectral and hue fusion data. Therefore, considering the comprehensive performance of classification accuracy, feature utilization capability and deployment feasibility, the 1D-CNN model is selected as the base model for subsequent optimization.

Ablation experiments

As shown in Table 8, the 1D-CBAMCNN model based on 205-dimensional spectral-hue multimodal features achieved comprehensively superior performance to the baseline 1D-CNN model in the 5-class tomato ripeness classification task, fully verifying the effectiveness of the CBAM attention mechanism for feature optimization; the overall accuracy of 1D-CBAMCNN reached 95.18%, representing an increase of 1.98% points compared with the 93.20% of 1D-CNN, and its macro-average and weighted-average precision, recall, and F1-score were also optimized synchronously, rising from 93.56%, 93.20%, and 93.38% of 1D-CNN to 95.18%, 95.18%, and 95.17%, respectively, which demonstrated the capability of the attention mechanism to accurately capture the global correlations of multimodal features. In terms of class-level performance, 1D-CBAMCNN exhibited particularly significant optimization effects for GRS and CCS, where 1D-CNN showed weak performance; the precision of GRS increased from 90.38% to 94.23%, and the recall of CCS rose from 84.00% to 89.58%, effectively reducing

misjudgments of early ripeness stages, while for high-discrimination categories from ERR to LRR, the model further enhanced feature stability while maintaining nearly perfect performance. In terms of computational efficiency, the FLOPs (18.76 M) and single-sample inference time (2.35 ms) of 1D-CBAMCNN only increased by 17.0% and 15.2% respectively compared with 1D-CNN, without causing a substantial surge in computational burden and still meeting the requirements of real-time detection in agricultural field scenarios. In summary, through channel and spatial attention weighting, 1D-CBAMCNN not only enhanced the key information in the spectral-hue fused features (such as chlorophyll absorption spectral bands and hue gradient features) but also suppressed redundant noise; it achieved a better balance among classification accuracy, category adaptability, and engineering practicability, providing a more reliable technical solution for high-precision non-destructive detection of tomato ripeness. Figure 9(a) shows the confusion matrix of classification results for tomatoes at different ripeness stages based on the 1D-CNN model. Figure 9(b) shows the confusion matrix of the 1D-CBAMCNN model. A paired t-test was conducted to compare the per-class accuracy of the two models, revealing that the 1D-CBAMCNN model achieved a statistically significant improvement over the 1D-CNN model ($t=3.333$, $p=0.031<0.05$). This result confirms that the integration of the CBAM attention mechanism effectively enhances the model's ability to identify tomatoes at different ripeness stages (Fig. 9).

Interpretability analysis based on SHAP

To address the “black box” characteristics of the 1D-CBAMCNN model, the SHAP method is used to analyze the feature importance of the classification results of

Table 8 Classification results of tomato ripeness using different models with multimodal spectral-hue features

Model	Number of feature wavelengths	Level	Precision/%	Recall/%	F1-Score/%	Accuracy/%	FLOPs/M	Average Inference Time per Sample/ms
1D-CNN	205	GRS	90.38	94.00	92.15	93.20	16.03	2.04
		CCS	87.50	84.00	85.71			
		ERR	93.75	90.00	91.84			
		MRR	98.00	98.00	98.00			
		LRR	98.04	100.00	99.01			
		Macro Avg	93.53	93.20	93.34			
		Weighted Avg	93.56	93.20	93.38			
1D-CBAMCNN	205	GRS	94.23	96.08	95.15	95.18	18.76	2.35
		CCS	89.58	89.58	89.58			
		ERR	95.83	92.00	93.88			
		MRR	98.00	98.00	98.00			
		LRR	98.04	100.00	99.01			
		Macro Avg	95.14	95.13	95.12			
		Weighted Avg	95.18	95.18	95.17			

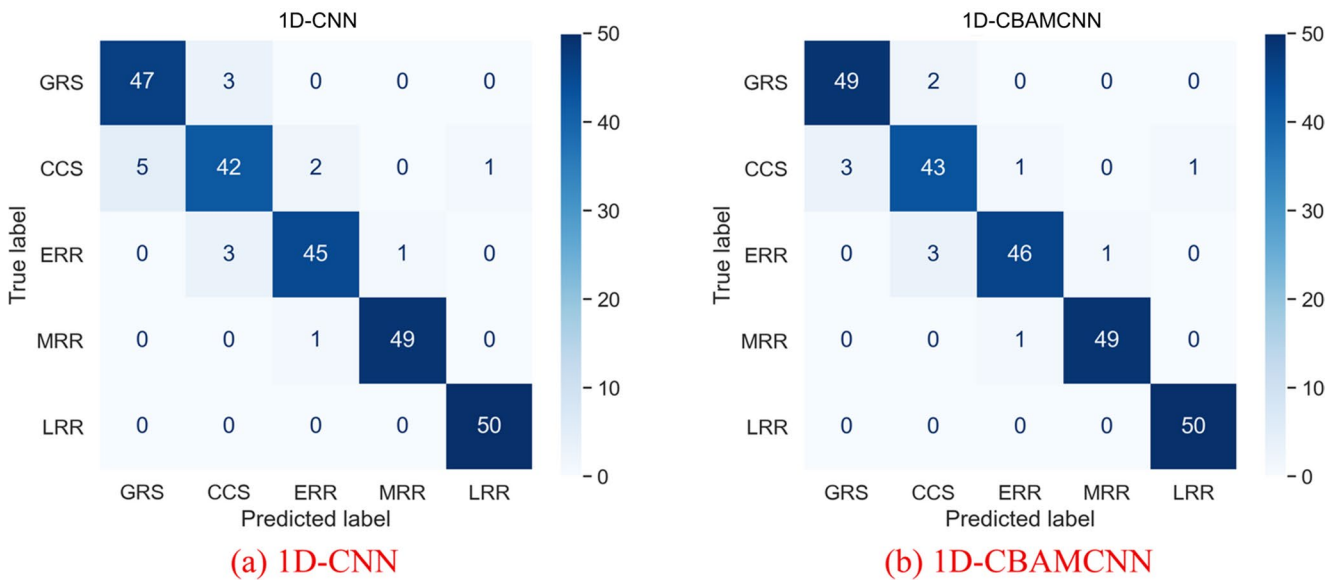


Fig. 9 Confusion matrix. (a) Classification results of 1D-CNN model; (b) Classification results of 1D-CBAMCNN model.

the five maturity categories in the test set of multimodal data samples after spectral-color fusion. The SHAP value is based on the principle of quantifying the contribution of each feature to the decision-making of the model based on the SHAPley value of game theory, and we quantify the contribution of each feature to the model's decision-making and its positive or negative impacts by calculating their SHAPley values, so as to quantify the role of different features on the model decision.

Figure 10 shows the feature contribution plots for the five maturity categories of tomato, showing the importance distribution of the 10 important features. Where the horizontal axis is the SHAP value and the vertical axis is the wavelength points, sorted from top to bottom in order of importance, each point being an actual sample in the test set, and the color of each point ranging from yellow to violet indicating that the specific wavelength point corresponds to a spectral intensity value from high to low.

Figure 10(a) shows the characteristic contribution map of the green ripening category. By analyzing the distribution of SHAP values, it can be found that the characteristic wavelengths such as HSV-H, 525 nm, 502 nm, 508 nm, etc. have a large positive influence on the green ripening category, which is manifested by the fact that the yellow dots are concentrated in the region larger than 0—these wavelengths in the 500–600 nm green light band confirm the high chlorophyll content and high reflectivity of immature tomatoes. On the contrary, the characteristic wavelengths such as 721 nm, 572 nm, etc. have a negative influence on the green ripening category, which is manifested by the fact that the yellow points are concentrated in the region

less than 0, consistent with the relatively low near-infrared reflectivity of green-ripening fruits. Figure 10(b) shows the characteristic contribution map of the color-turning phase category. By analyzing the distribution of SHAP values, it can be found that the characteristic wavelengths such as 578 nm, 584 nm, and 572 nm have a large positive influence on the color-turning phase category, while the characteristic wavelengths such as 682 nm, 688 nm, and 694 nm have a large negative influence on the color-turning phase category. The positive contribution of these 500–600 nm bands indicates the model's sensitivity to the relative reflectivity changes during this transitional ripening stage. Figure 10(c) shows the feature contribution map for the red-ripening pre-color category. By analyzing the distribution of SHAP values, it can be found that the wavelengths such as 420 nm, 406 nm, 605 nm have a greater positive influence on the red-ripening pre-color category, and the features such as HSV-H, 397 nm, 682 nm have a greater negative influence on the red-ripening pre-color category. Notably, the negative SHAP value of HSV-H aligns with the green-to-red color shift of maturing tomatoes, providing direct physiological evidence for the model's judgment. Figure 10(d) shows the contribution of features to the red-ripening intermediate category. By analyzing the distribution of SHAP values, it can be found that the wavelengths such as 605 nm, 406 nm, 599 nm have a large positive influence on the red-ripening intermediate category, while the wavelengths such as 409 nm, 415 nm, 628 nm have a negative influence on the red-ripening intermediate category. The positive contribution of 605 nm and 599 nm reflects the high reflectivity after chlorophyll degradation,

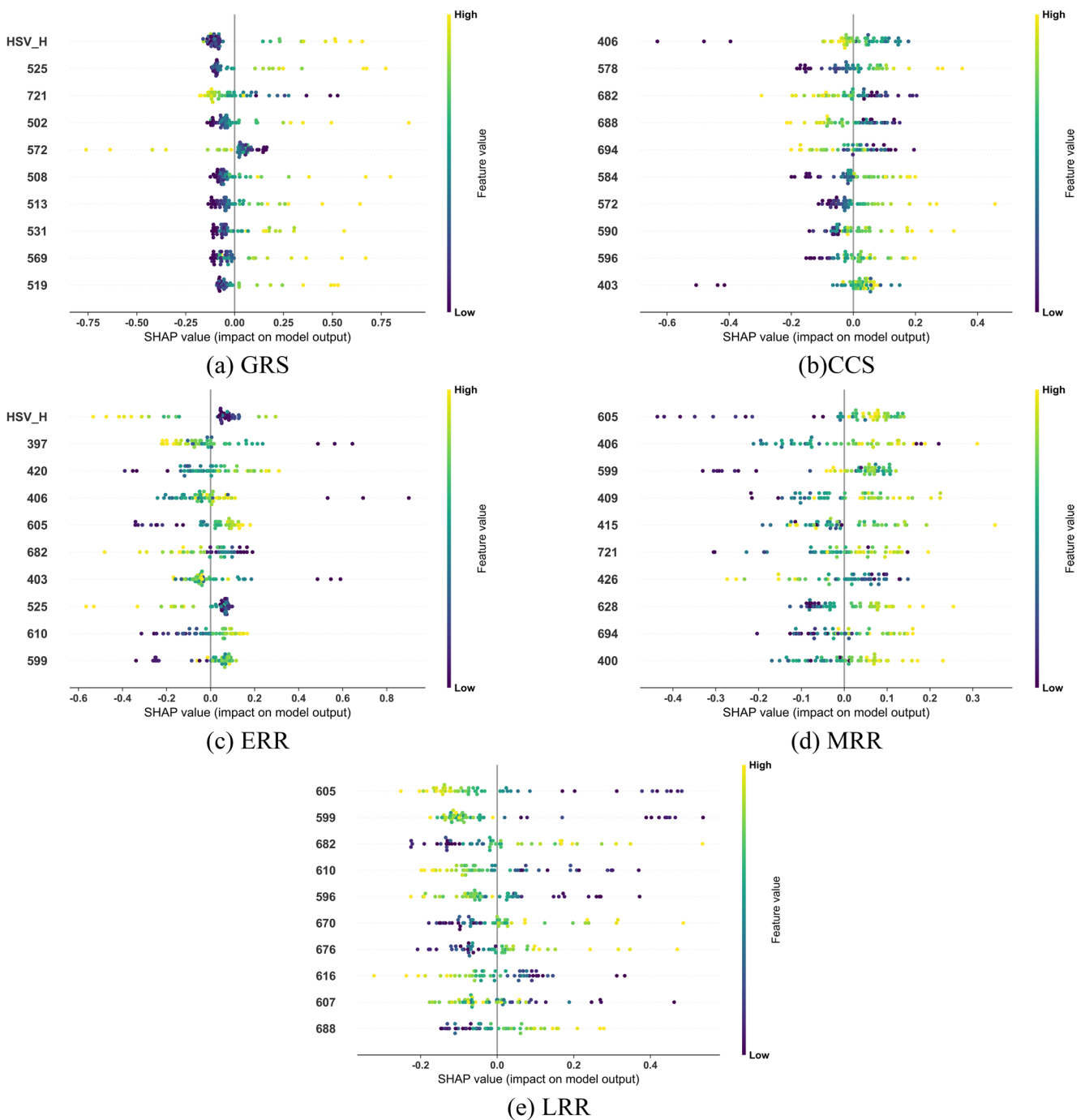


Fig. 10 Distribution of importance of SHAP features for different categories of samples.(a) GRS, (b)CCS, (c) ERR, (d) MRR, (e) LRR

while the negative 415 nm response matches the strong absorption of carotenoids like lycopene in the 400–500 nm band. Figure 10(e) shows the characteristic contribution of the late red-ripening category. By analyzing the distribution of SHAP values, it can be found that wavelengths such as 682 nm, 670 nm, and 676 nm have a large positive

influence on the late red-ripening category, while wavelengths such as 605 nm, 599 nm, and 610 nm have a negative influence on the late red-ripening category. The positive 670–682 nm bands correspond to the peak reflectivity near the chlorophyll absorption valley, a key spectral marker of fully ripe tomatoes with negligible chlorophyll.

Discussion

In this study, feature wavelength screening significantly improved the prediction accuracy of TSS and firmness, indicating that redundant information in raw spectra can interfere with model performance. Spectral preprocessing and feature screening effectively retain key wavelengths and reduce noise interference, which are important steps in optimizing model accuracy. Multimodal fusion works as hyperspectral data and HSV hue H-values complement each other. For example, in the green-to-color-changing transition, hyperspectral data alone barely distinguishes subtle internal changes, but HSV hue H-values identify initial color changes. The 1D-CNN processed fused data well, and the 1D-CBAMCNN with CBAM attention further boosted performance by focusing on core features. Though the basic 1D-CNN architecture is not novel, the CBAM module is strategically integrated to target the spectral characteristics of tomato ripening, which is the key to achieving superior results with a conventional model. It assigns high weights to 750–950 nm firmness-related wavelengths and key HSV hue H-values, while ignoring irrelevant information. This targeted optimization makes our model outperform existing studies that use the same basic 1D-CNN for tomato ripeness detection [55]. The SHAP method enables model interpretability by clarifying feature roles in each stage. Key wavelengths align with tomato ripening physiology: 450–650 nm dominates the green ripe stage, and 750–950 nm dominates the red ripe stage, proving the model reliable.

This study has achieved preliminary results in non-destructive tomato ripeness detection, but there is still room for improvement due to resource constraints. First, we only used Jingcai 8 tomatoes as samples. This variety is a locally dominant fresh-eating tomato with high market share and strong representativeness, but the lack of multi-varietal validation limits the generalization ability of the proposed method. Future work will expand the sample scope to investigate spectral differences across tomato varieties and producing areas to improve the method's universality. Second, with approximately 50 samples per ripening stage, the relatively small dataset poses a potential overfitting risk. Although we mitigated this issue by optimizing the model structure and implementing strict cross-validation, the generalization performance of the model on unseen samples still needs to be verified with larger-scale datasets. Third, the practical deployment of hyperspectral imaging technology in real farm scenarios faces multiple challenges: hyperspectral devices are generally bulky and costly, which hinders on-site rapid detection; complex and variable field lighting conditions can easily interfere with the stability of spectral data collection; and issues such as plant occlusion and irregular fruit postures may affect the acquisition of

valid spectral information [25, 37]. These practical obstacles need to be addressed in subsequent research.

Conclusion

This study focuses on the accurate grading of tomato ripeness based on hyperspectral imaging technology combined with multimodal fusion, with the key conclusions drawn as follows:

1. The optimal model configuration is determined as SNV preprocessing+firmness-based feature wavelength selection+hyperspectral-HSV H-value fusion+1D-CBAMCNN.
2. This optimal configuration significantly improves the prediction accuracy of soluble solids and firmness, as well as the performance of tomato ripeness grading. It outperforms the full-wavelength model, single-modal models, and the basic 1D-CNN model without the CBAM attention module.
3. SHAP analysis clarifies the role of each feature in different tomato ripening stages, which confirms the interpretable decision-making of the proposed model.

Future research will expand tomato datasets with diverse varieties and origins, and adopt domain adaptation techniques such as transfer learning to enhance the model's adaptability to complex agricultural environments. We will also explore lightweight model deployment technologies including model pruning and quantization to facilitate on-site rapid detection. Looking forward, we aim to integrate the proposed method into portable hyperspectral devices to realize real-time and on-site assessment of tomato ripeness, and further explore its combination with UAV-based multispectral imaging systems. This will enable the construction of a comprehensive detection system integrating large-area aerial screening and ground precise validation, thus providing technical support for field management in smart agriculture. Additionally, we will further deepen the mechanistic interpretation of key spectral features associated with tomato ripening physiology.

Acknowledgements The authors would like to thank the helpful comments and suggestions of the reviewers, which have improved the presentation. This work was supported by the National Natural Science Foundation of China (NO. 62176261, NO. 62376272).

Author contribution Qin Zhou: Design, Methodology, Experimentation, Theoretical analysis, Writing-original draft. Shiai Zhou: Data curation, Writing – review & editing. Xiaohuan Li: Data curation. Yiding Zhang: Supervision, Writing – review & editing. Lingxian Zhang: Funding acquisition, Conceptualization, Supervision, Writing – review & editing.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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